
Genetic Algorithms

Optimization inspired by natural evolution

Selection -> Crossover -> Mutation -> Next Generation

Solution Representation

Each candidate solution is encoded as a chromosome



Fitness Function

$f(x)$ = number of 1s in the genome $\rightarrow$ higher = better

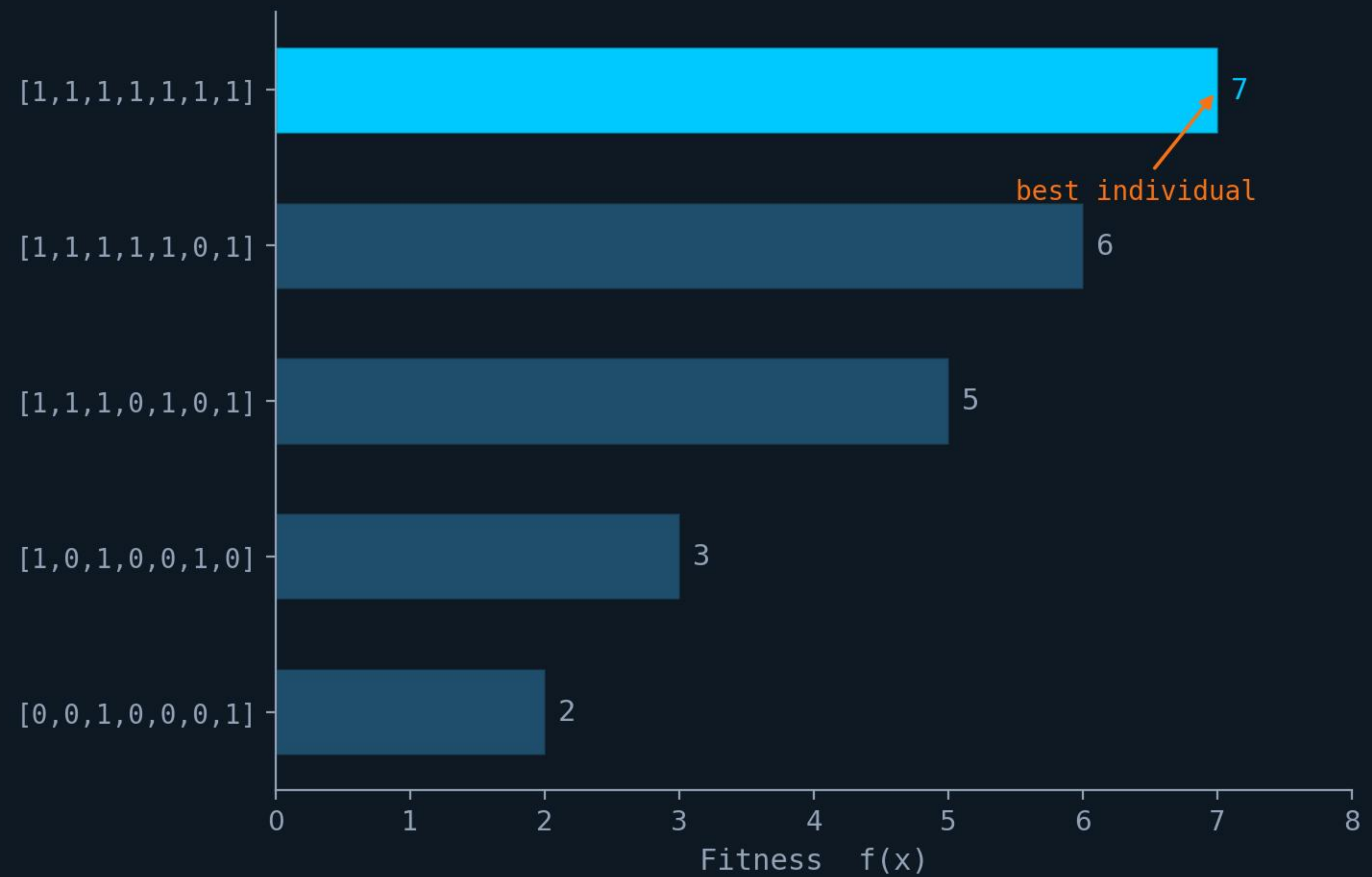
genome
[1,0,1,1,0,0,1]



fitness
function $f(x)$



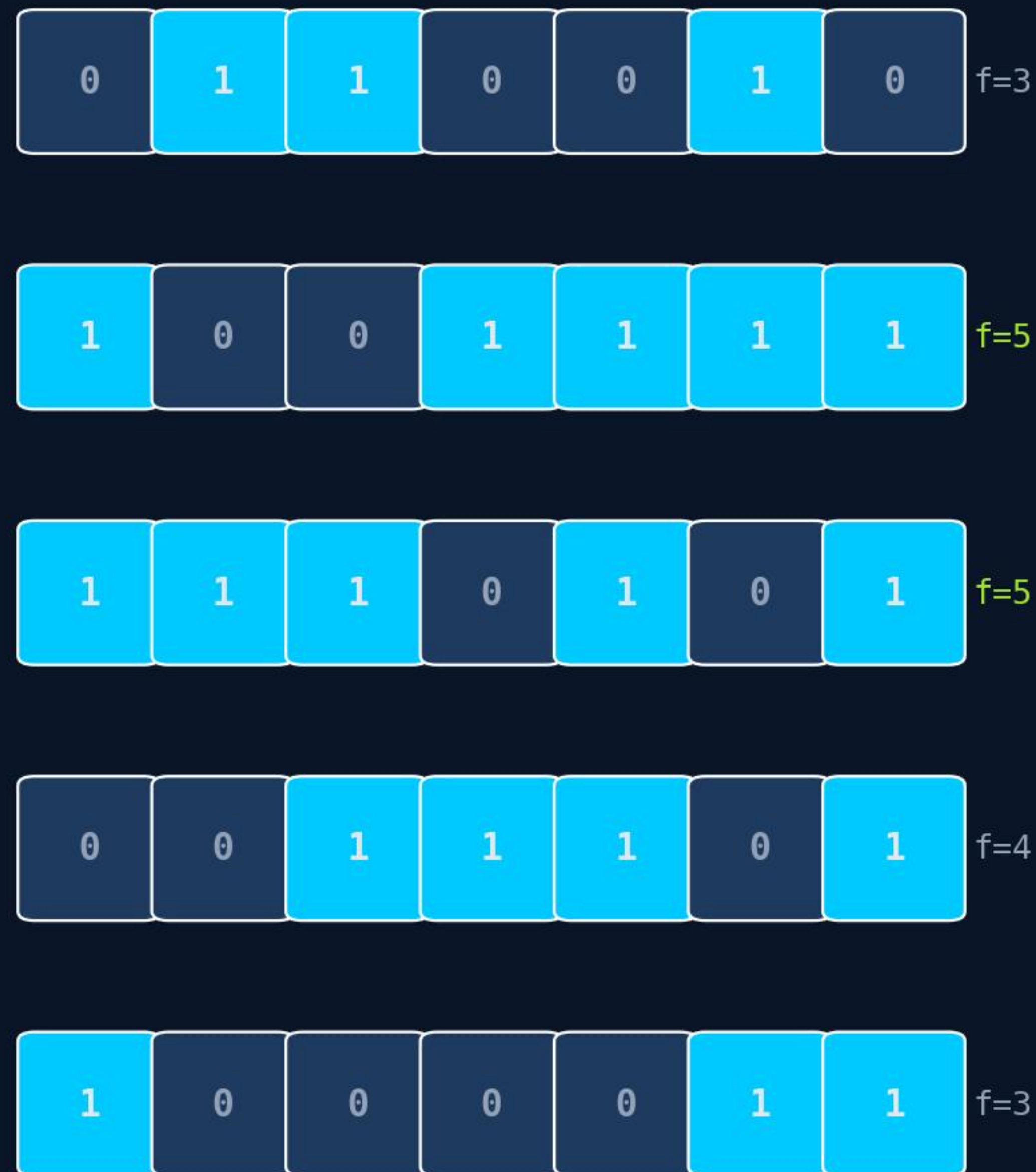
score
 $f = 4$



Population & Generations

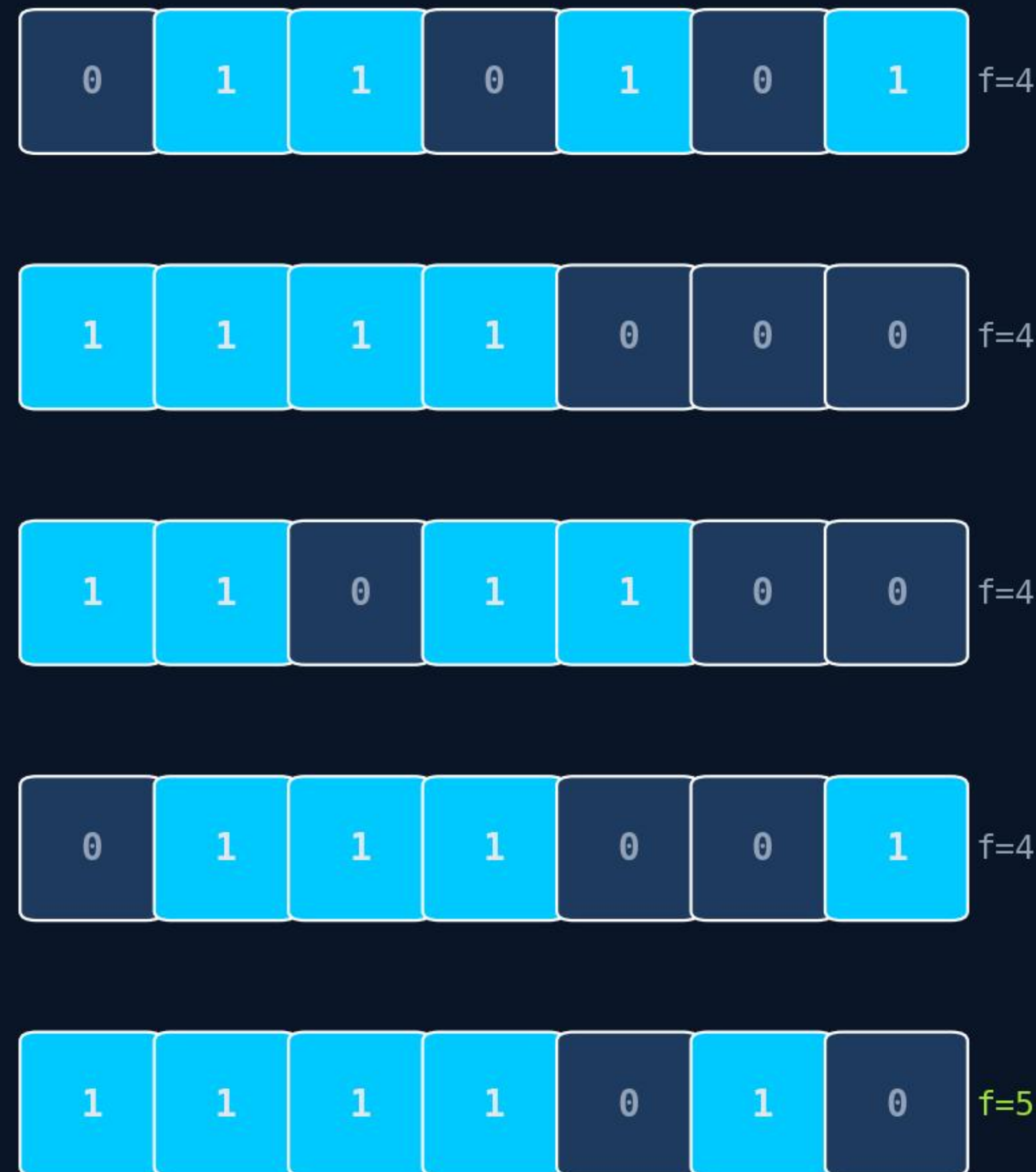
A population evolves over many generations

Generation 1



evolve

Generation 2



evolve

Generation 3



Selection - Tournament

Pick k random individuals $\rightarrow$ the fittest one becomes a parent

1	1	1	1	1	1	1
---	---	---	---	---	---	---

$f = 7$

0	0	0	0	1	1	0
---	---	---	---	---	---	---

$f = 2$

0	1	0	1	0	0	1
---	---	---	---	---	---	---

$f = 3$

0	0	0	1	0	1	0
---	---	---	---	---	---	---

$f = 2$

[WINNER]

0	1	1	1	1	1	1
---	---	---	---	---	---	---

$f = 6$

1	1	1	0	1	0	0
---	---	---	---	---	---	---

$f = 4$



winner



tournament pool



not selected

Reproduction - Crossover

Combine two parents to create offspring



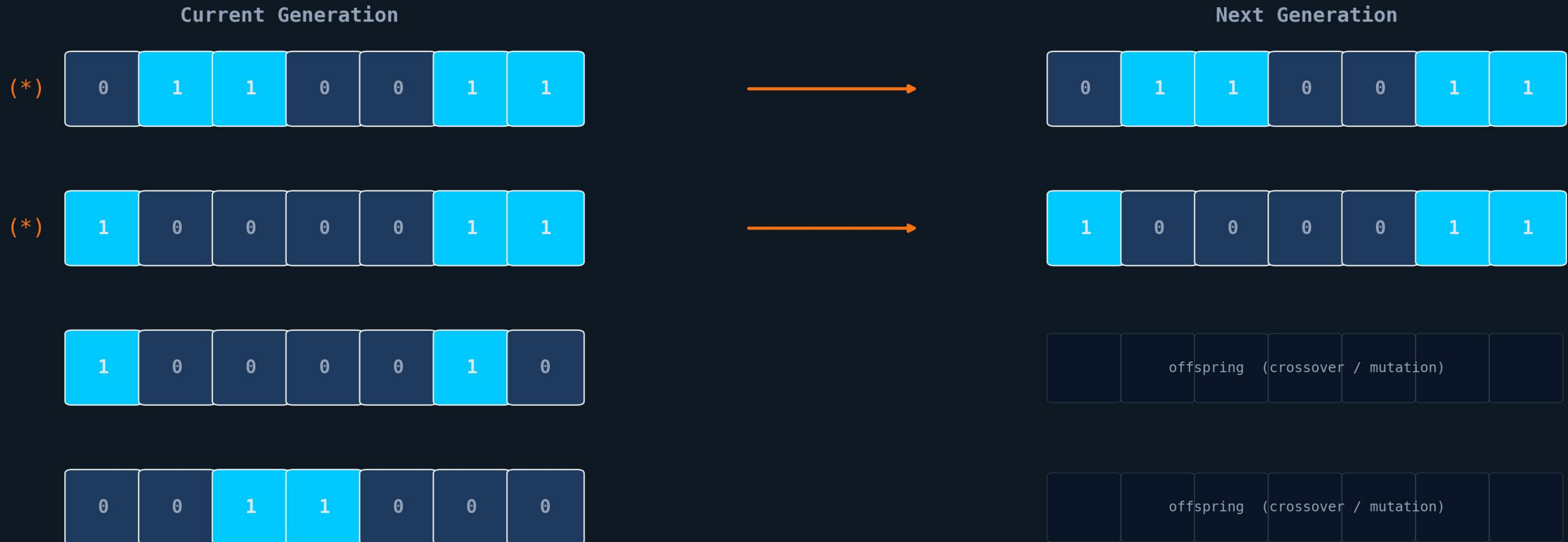
Reproduction - Mutation

A random gene is flipped -> maintains diversity



Elitism

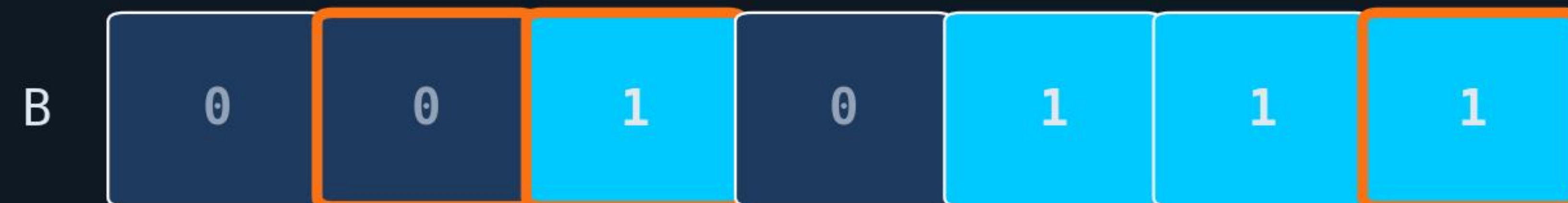
The best N individuals survive unchanged into the next generation



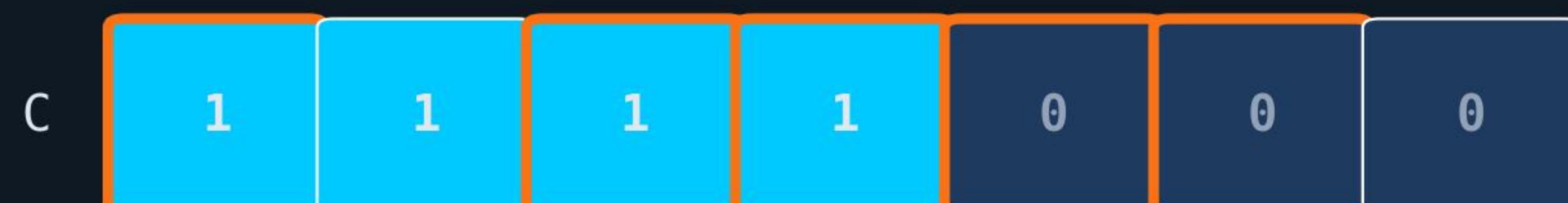
Speciation & Fitness Sharing

Group similar individuals into species -> encourage diversity

Hamming Distance



$H(A,B) = 3$ -> same species (threshold = 3)



$H(A,C) = 5$ -> different species

Fitness Sharing

Species 1



$f = 6$

$|s| = 4$

$f' = 1.5$

Species 2



$f = 5$

$|s| = 1$

$f' = 5.0$

$$f'(i) = f(i) / |s(i)|$$

Genetic Algorithm - Overview

